

Household Services Application - Project Report

AUTHOR Name : Deepak Kumar

Email ID : 22f3000107@ds.study.iitm.ac.in

1. Introduction

- **Project Title:** Household Services Application - V1
- **Project Summary:** This project aims to create a multi-user household services platform where users can either request or provide household services. Three distinct user roles—Admin, Service Professional, and Customer—enable a streamlined process for service management.
- **Objectives:**
 - To facilitate an efficient booking and management system for household services.
 - To implement a role-based access system for secure and personalized experience.

2. Technology Used

- **Flask:** Backend framework for building the web application.
- **SQLAlchemy:** ORM (Object-Relational Mapping) tool for database interactions.
- **SQLite :** Database management system for storing application data.
- **HTML/CSS/JavaScript:** Frontend technologies for user interface design and interactivity.
- **Flask-Login:** Extension for managing user sessions and authentication.
- **Datetime:** Python library for handling date and time operations.

Additional Libraries:

- Werkzeug (for password hashing and authentication)
- Flask-WTF (for form handling)
- Flask-Login (for session management)

3. System Architecture

- **Database Models:**
 - **User Model:** Stores user information along with their role (Admin, Professional, Customer).
 - **Service Model:** Contains service details such as type, description, and available professionals.
 - **Role Model:** Defines the roles and access privileges.
- **Role-Based Access Control (RBAC):** Implements access restrictions based on roles:
 - **Admin:** Manages services and user accounts.
 - **Service Professional:** Can view and manage their accepted service requests.
 - **Customer:** Can request services, view order status, and manage their bookings.

4. Features and Functionalities

- **User Authentication and Role-Based Access:**
 - Built using Flask-Security and SQLAlchemy, users are directed to role-specific dashboards upon login.
 - Different dashboards for each role include functionalities tailored to their

respective needs.

- **Service Request System:**

- Customers can book services, and Service Professionals can accept or reject requests.

- Admins can manage available services and user accounts.

- **Dynamic Signup Fields:**

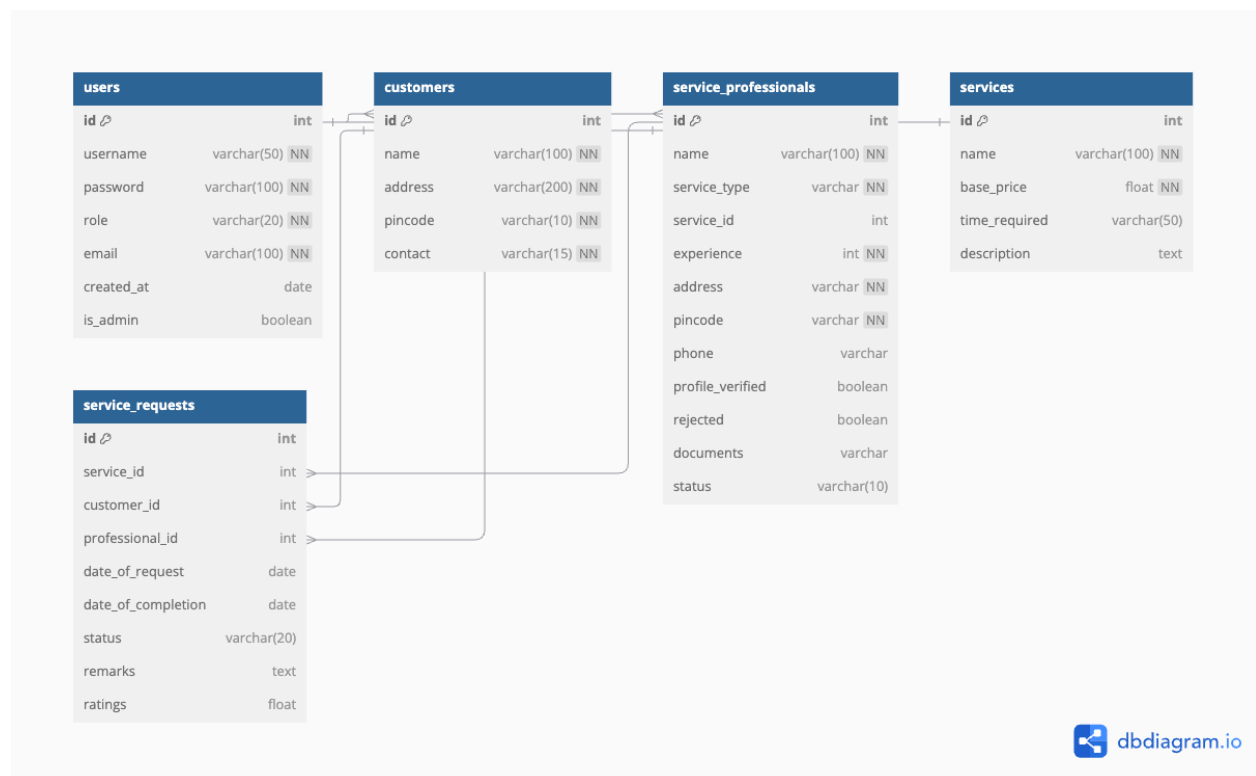
- The signup page includes role-specific fields, dynamically displayed based on the user's selected role.

- **Error Handling and Validation:**

- Ensures users receive appropriate feedback on failed actions and error messages are logged for debugging.

5.DB Schema Design

The diagram includes five main tables: **User**, **Customer**, **ServiceProfessional**, **Service**, and **ServiceRequest**, along with their attributes and relationships.



Relationships

1. A **User** can be either a **Customer** or a **ServiceProfessional**, represented by one-to-one relationships.
2. A **Customer** can make multiple **ServiceRequests** (One-to-Many relationship).
3. A **ServiceProfessional** can handle multiple **ServiceRequests** (One-to-Many relationship).
4. A **Service** can have multiple **ServiceRequests** and multiple associated **ServiceProfessionals**.

6.Conclusion

- **Outcomes:** This project successfully creates an interactive platform for household services, providing specific functionality for admins, professionals, and customers.

- **Future Work:** Potential enhancements include integrating payment processing and adding a review system for professionals.

7. Presentation(video link)

https://drive.google.com/file/d/1-D-DCD_60kWwTBcbKyzEGKV9PKkaw7DjF/view?usp=drive_link